AI-Powered HR Screening and Document Retrieval System Using RAG

HIREFLOW

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GitHub

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OBJECTIVE / ABSTRACT

Hireflow is a self-hostable, AI-powered HR screening and document-retrieval platform built on Retrieval-Augmented Generation (RAG).

It lets HR teams ingest heterogeneous documents, search and question them in plain language, and automatically screen and rank candidates against job criteria, with every answer traceable to its source. Key objectives:

- Ingest and OCR PDF, Word, and scanned documents (PyMuPDF and Tesseract), with over 95% text accuracy on printed material.
- Semantic hybrid search over a ChromaDB vector store.
- Grounded, source-cited answers via RAG (Anthropic Claude), refusing out-of-scope questions.
- Transparent, weighted resume-to-job scoring.
- Automated candidate intake through Gmail synchronisation.

INTRODUCTION

Organisations accumulate large volumes of unstructured documents, and HR teams spend the majority of their time manually reviewing resumes, on average around 23 hours to fill a single position.

Keyword search matches literal tokens rather than meaning, so a query for “machine learning engineer” misses a strong candidate who wrote “ML researcher”. Hireflow replaces manual review and rigid keyword filters with semantic retrieval and grounded question-answering, unifying document intake, search, and candidate screening in one coherent platform.

METHODOLOGY

Two pipelines drive the system.

Ingestion (asynchronous, on a Celery worker): each document is extracted, classified, chunked, contextualised, embedded with the bge-small-en-v1.5 model, and indexed into ChromaDB. Text and metadata are stored in PostgreSQL and the original files in MinIO.

Retrieval and RAG (per request): the query is parsed and normalised; vector, lexical (tsvector), metadata, and fuzzy retrieval then run and are combined by Reciprocal-Rank Fusion. An intent-aware prompt is composed and a source-cited answer is streamed from Claude.

The backend is a strictly layered FastAPI application (API, Service, Domain, and Adapter layers) with Protocol-based, swappable providers, so business rules stay testable and infrastructure remains replaceable. The frontend is a React 19 single-page application.

DESIGNED PROJECT

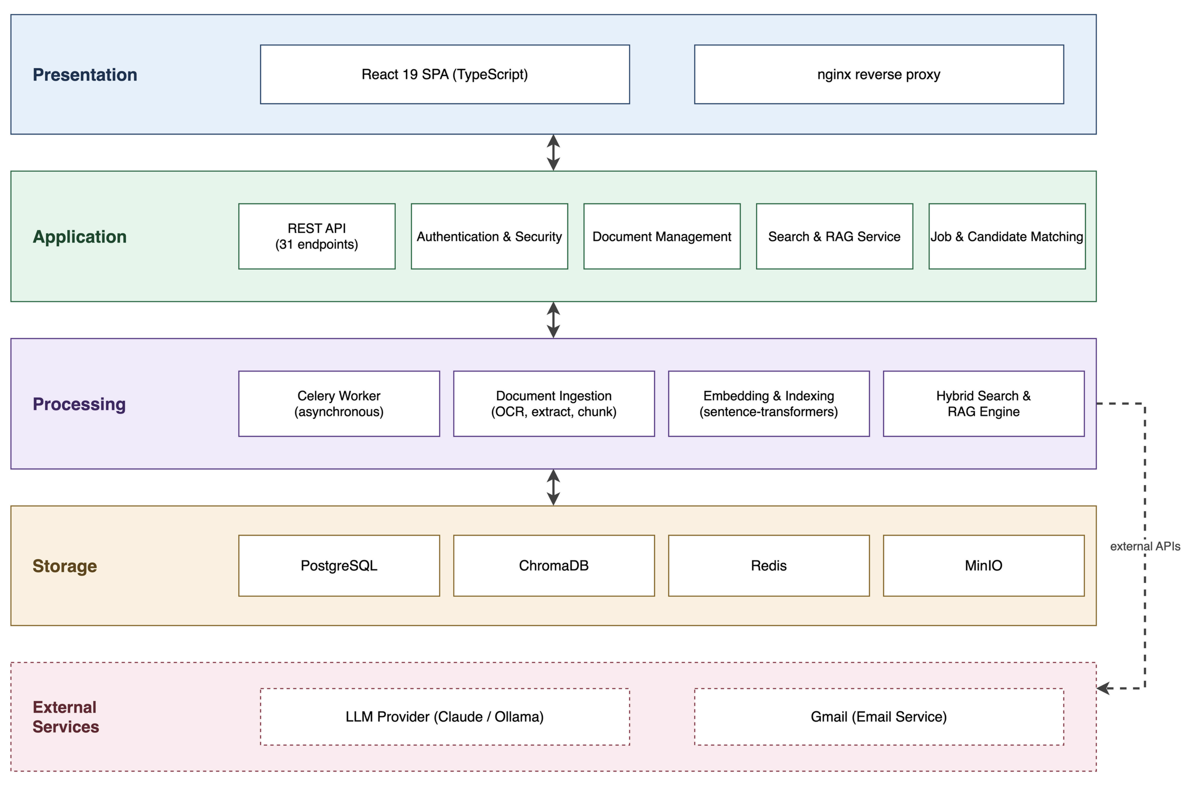

Figure 1: Hireflow layered system architecture. The Presentation, Application, Processing, and Storage layers sit over the data stores, with external LLM and Gmail services.

RESULTS

72 ms Hybrid search, median latency	**947 ms** RAG time to first token	**1.5 s** Full streamed RAG answer
0.57 ms Per-chunk embedding on GPU	**>95%** OCR text accuracy, printed	**31** Working REST API endpoints

CONCLUSION

Hireflow meets its four objectives: document management, natural-language search, transparent resume screening, and automated intake, in one coherent, self-hostable system. Every generated answer is source-attributed, and out-of-scope questions are refused rather than hallucinated.

Future work includes login rate-limiting, refresh-token revocation on password change, and horizontal scaling of the worker pool.

REFERENCES

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3. Vaswani et al. (2017). Attention Is All You Need. NeurIPS 30.
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TECHNOLOGIES USED

Python | **FastAPI** | **React 19** | **TypeScript** | **PostgreSQL 15** | **ChromaDB** | **Redis** | **Celery** | **MinIO** | **Anthropic Claude** | **sentence-transformers** | **Tesseract OCR** | **PyMuPDF** | **Docker**